

## Automata, Logic and Games HW 2

### Assignment 4

1. Yes. Let  $A = \{0, 1\}$ ,  $t(w) = |w| \bmod 2$ , and  $L = \{t\}$  - regular, since  $t$  is regular.  $t$  is obviously not constant, but is dense, as  $t(a_1 a_2 w) = t(w)$ .
2. No. Using a theorem from the slides, if a non-deterministic parity automaton accepts some tree, it must accept some regular tree. A regular tree can only have finitely many distinct subtrees, which means it is not super dense.
3. No. Assuming  $|A| \geq 2$ , there are uncountably many trees. but since any specific tree has countably many nodes, it can only have countably many subtrees (each subtree has a root in some node), therefore this tree can't have all possible trees as it's subtrees by counting argument.

### Assignment 6

1. No. Let  $L$  be regular, recognized by  $\mathbf{A}$  with a state set  $\mathbf{Q}$

Observation 1. If  $t, t' \in L$  have accepting runs  $\rho, \rho'$  and there are nodes  $u, v$  with  $\rho(u) = \rho'(v)$ , then the tree obtained from  $t$  by replacing the subtree  $t \upharpoonright_u$  with  $t' \upharpoonright_v$  is still in  $L$ .

Observation 2. If  $L$  is not empty, it contains a regular tree.

Observation 3. A regular tree has only finitely many distinct subtrees.

Assume for contradiction that  $L$  is infinite and every tree in  $L$  is dense. Pick a regular tree  $t \in L$  with an accepting run  $\rho$ . Choose a node  $u = p\sigma \neq \epsilon$  in  $t$  and let  $q = \rho(u)$ . Let

$$L_q := \{s \mid \mathbf{A} \text{ accepts } s \text{ from state } q\}.$$

Since  $L$  is infinite, we can choose such  $q$  so that  $L_q$  is infinite.

Since  $t$  has only finitely many subtrees and  $L_q$  is infinite, pick  $s \in L_q$  that it is not equal to to any subtree of  $t$ . Now insert  $s$  into  $t$  at  $u$ . The resulting tree  $r$  is still in  $L$ .

Since  $s$  is not equal to any subtree of  $t$ , it does not occur as a subtree of the sibling of  $u$ , meaning that  $r$  is not dense.

## Assignment 7

1. No. Let  $A = \{0, 1\}$ . Take a word  $m$  s.t.  $\{m\}$  is not  $\omega$ -regular, for example  $m_i = 1$  iff  $i$  is prime. Let  $t(u) = w_{|u|}$ . Trivially,  $\text{walks}(t) = \{w\}$  which is not  $\omega$ -regular
2. Yes. Take  $\mathbf{A}$  recognizing  $L$ , and it's set state  $Q$ . We will show that the language of walks is recognized by a parity word automaton  $\mathbf{B}$ .  $\mathbf{B}$  has the same set of states and the beginning state as  $\mathbf{A}$ . It reads a word  $w$  and simulates  $\mathbf{A}$  along a non-deterministically guessed branch. If there is no valid run in  $\mathbf{A}$  or the other subtree of the guessed branch can't be accepted by  $\mathbf{A}$ , it simply rejects.
3. Yes.  $L$  contains a regular tree  $t$  (as shown previously), a regular tree has finitely many distinct subtrees, and we can think of moving through a regular tree as walking on a graph of subtrees, created the following way: the vertices in this graph are distinct subtrees, each vertex has two edges, for two of the subtrees that begin in children root of the node of this specific "subtree type". We can have a Buchi automaton that guesses walks on this finite graph, making the language  $\omega$ -regular.